DEPARTMENT OF MATHEMATICS

MAT122 INTRODUCTORY MATHEMATICS II

WEEK 6: LECTURE NOTE

DIFFERENTIATION continued......

Plane Curves, Parametric Equations and Calculus

Plane Curves and Parametric Equations

Definition 1 (**Definition of a Plane Curve**). *If f and g are continuous functions of t on an interval I, then the equations*

$$x = f(t) \qquad \text{and} \qquad y = g(t)$$

*are **parametric equations** and t is **the parameter**. The set of points (x, y) obtained as t varies over the interval I is **the graph of the parametric equations**. Taken together, the parametric equations and the graph are a **plane curve**, denoted by C.*

When sketching a curve represented by a set of parametric equations, you can plot points in the xy-plane. Each set of coordinates (x, y) is determined from a value chosen for the parameter t. By plotting the resulting points in order of increasing values of t the curve is traced out in a specific direction. This is called **the orientation of the curve**.

Example 2. ***Sketching a Curve*** *Sketch the curve described by the parametric equations*

$$x = f(t) = t^2 - 4 \qquad \text{and} \qquad y = g(t) = \frac{t}{2}, \qquad \text{where} \quad -2 \leq t \leq 3.$$

Solution.
Solution in class.

Eliminating the Parameter

Finding a rectangular equation that represents the graph of a set of parametric equations is called **eliminating the parameter**. Follow the following steps to eliminate parameters:

STEP 1. Parametric equations

$$x = f(t) = t^2 - 4 \qquad \text{and} \qquad y = g(t) = \frac{t}{2}, \qquad \text{where} \quad -2 \leq t \leq 3.$$

STEP 2. Solve for t in one equation (simpler one).

$$t = 2y.$$

STEP 3. Substitute into second equation.

$$x = (2y)^2 - 4.$$

STEP 4. Rectangular equation

$$x = 4y^2 - 4.$$

Once you have eliminated the parameter, you can recognize that the equation $x = 4y^2 - 4$ represents a parabola with a horizontal axis and vertex at (-4,0) as shown in the figure above.

The range of and implied by the parametric equations may be altered by the change to rectangular form. In such instances, the domain of the rectangular equation must be adjusted so that its graph matches the graph of the parametric equations.

Example 3. ***Adjusting the Domain***
Sketch the curve represented by the equations

$$x = \frac{1}{\sqrt{t+1}} \qquad \text{and} \qquad y = \frac{t}{t+1}, \quad t > -1$$

by eliminating the parameter and adjusting the domain of the resulting rectangular equation.
Solution. *Solution in class.*

Example 4. ***Using Trigonometry to Eliminate a Parameter***
Sketch the curve represented by

$$x = 3\cos\theta \qquad \text{and} \qquad y = 4\sin\theta, \quad 0 \le \theta \le 2\pi$$

by eliminating the parameter and finding the corresponding rectangular equation.
Solution. *Solution in class.*

Finding parametric equations

We have discussed the techniques for sketching the graph of equations represented by a set of parametric equations. We will now investigate the reverse problem. How can you determine a set of parametric equations for a given graph or a given physical description? The main issue here is that such a representation is **not** unique.

Example 5. ***Finding Parametric Equations for a Given Graph***
Find a set of parametric equations that represents the graph of $y = 1 - x^2$, using each of the following parameters.

a. t *b.* The slope $m = \dfrac{dy}{dx}$ at the point

Solution
Solution in class.

Example 6. ***Parametric Equations for a Cycloid***
Determine the curve traced by a point on the circumference of a circle of radius a rolling along a straight line in a plane. Such a curve is called a ***cycloid****.*

Solution. *Let the parameter θ be the measure of the circle's rotation, and let the point $P(x, y)$ begin at the origin. When $\theta = 0$, P is at the origin. When $\theta = \pi$, P is at a maximum point $(\pi a, 2a)$. When $\theta = 2\pi$, P is back on the $x-$axis at $(2\pi a, 0)$. The parametric equations of the cycloid are:*

$$x = a(\theta - \sin\theta) \qquad \text{and} \qquad y = a(1 - \cos\theta).$$

Definition 7. ***Definition of a Smooth Curve.***
A curve represented by $x = f(t)$ and $y = g(t)$ on an interval I is called ***smooth*** *when f' and g' are continuous on I and not simultaneously 0, except possibly at the endpoints of I. The curve is called* ***piecewise smooth*** *when it is smooth on each subinterval of some partition of I.*

Parametric Equations and Calculus

Theorem 8 (**Parametric Form of the Derivative**). *If a smooth curve C is given by the equations*

$$x = f(t) \qquad \text{and} \qquad y = g(t),$$

then the slope of C at (x, y) is

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{g'(t)}{f'(t)}, \quad \frac{dx}{dt} \neq 0.$$

Example 9. *Find $\frac{dy}{dx}$ for the curve given by $x = \sin t$ and $y = \cos t$.*
Solution.
Solution in class.

Example 10. *For the curve given by*

$$x = \sqrt{t} \qquad \text{and} \qquad y = \frac{1}{4}[t^2 - 4], \quad t \geq 0$$

find the slope at the point $(2, 3)$.
Solution.
Solution in class.

Example 11. *The prolate cycloid is given by*

$$x = 2t - \pi \sin t \qquad \text{and} \qquad y = 2 - \pi \cos t.$$

Show that the prolate cycloid crosses itself at the point $(0, 2)$. Find the equations of both tangent lines at this point.
Solution.
Solution in class.

Remark 12.

1. *If $\frac{dy}{dt} = 0$ and $\frac{dx}{dt} \neq 0$ when $t = t_0$, then the curve represented by $x = f(t)$ and $y = g(t)$ has a* ***horizontal tangent*** *at $(f(t_0), g(t_0))$.*

2. *If $\frac{dy}{dt} \neq 0$ and $\frac{dx}{dt} = 0$ when $t = t_0$, then the curve represented by $x = f(t)$ and $y = g(t)$ has a* ***vertical tangent*** *at $(f(t_0), g(t_0))$.*